IGCSE 0607 Practice Questions — E4.2 — Combined Questions & Markscheme

Gradient of linear graphs

Total Questions: 15 — **Total Marks:** 93 Marks

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  → **Paper 2 (Non-Calculator)** *(10 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

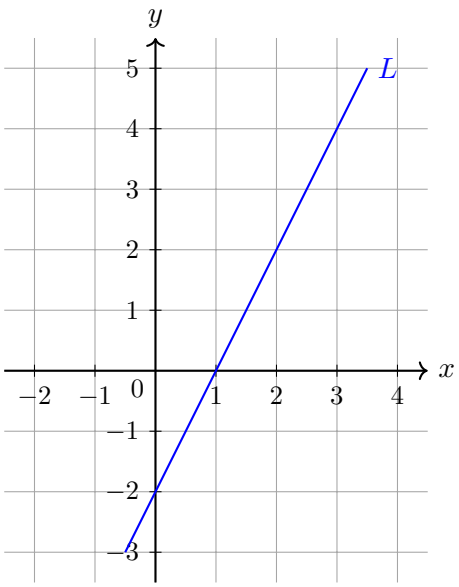
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1 Points A and B have coordinates $(2, -3)$ and $(6, 5)$ respectively.

Work out the gradient of the line AB.

..... **[2]**

2 The diagram shows a straight line L drawn on a coordinate grid.



(a) Find the gradient of line L.

..... **[2]**

(b) Write down the coordinates of the y-intercept of line L.

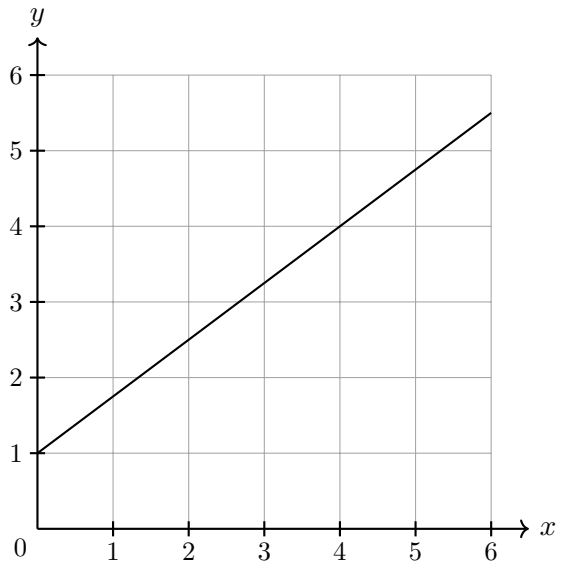
(..... , ...) **[1]**

3 The points A and B have coordinates $(-2, 8)$ and $(4, -4)$ respectively.

(a) Work out the gradient of the line AB.

..... **[2]**

(b) A straight line L is drawn on the grid.
Find the gradient of line L.



..... **[2]**

4 The points $A(-4, 7)$ and $B(2, -5)$ lie on a straight line.

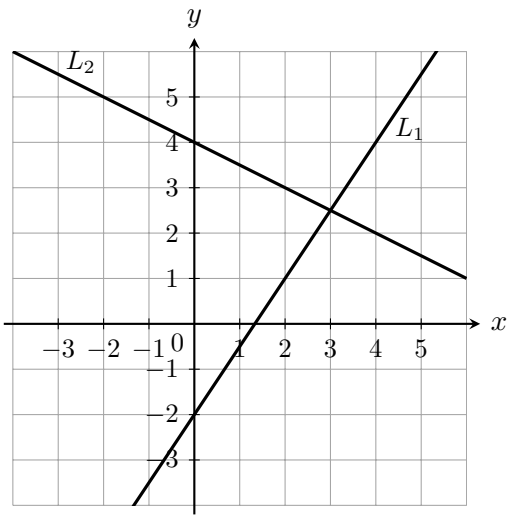
(a) Calculate the gradient of the line AB.

..... **[2]**

(b) Point C has coordinates $(0, c)$.
The gradient of the line AC is equal to the gradient of the line AB.
Find the value of c.

$c =$ ..... **[2]**

5 The diagram shows two straight lines, L_1 and L_2, drawn on a coordinate grid.



(a) Find the gradient of line L_1.

..... **[2]**

(b) Find the gradient of line L_2.

..... **[2]**

(c) A straight line L_3 passes through the points $P(-3, 7)$ and $Q(5, -9)$.
Calculate the gradient of line L_3.

..... **[2]**

6 The points $A(-2, 5)$, $B(4, k)$, and $C(10, -1)$ lie in a coordinate plane.

(a) Given that the gradient of the line segment AB is $-\frac{1}{2}$, calculate the value of k.

$k =$ ..... **[3]**

(b) Using your value of k from part (a), show that the points A, B, and C lie on the same straight line.

[2]

(c) Point D has coordinates $(p, 3p - 1)$.
Find the value of p for which the line AD has a gradient of 2.

$p =$ ..... **[3]**

7 **(a)** Point P has coordinates $(2a, 3a)$ and point Q has coordinates $(5a, a + 4)$.

Find an expression for the gradient of the line PQ in terms of a.
Give your answer in its simplest form.

..... **[2]**

(b) The line PQ is perpendicular to a line with gradient 3.
Work out the value of a.

$a =$ ..... **[3]**

(c) The points $A(-1, 2)$, $B(3, 8)$ and $C(c, 23)$ lie on a straight line.
Work out the value of c.

$c =$ ..... **[3]**

8 The line L_1 passes through the points $P(-2, 3a)$ and $Q(4, a+6)$.

(a) Find an expression for the gradient of L_1 in terms of a.
Give your answer in its simplest form.

..... **[3]**

(b) The line L_2 has equation $y = 3x - 5$.
Given that L_1 is parallel to L_2, find the value of a.

$a =$ ..... **[2]**

(c) Using your value of a, find the coordinates of the midpoint of PQ.

(..... , ...) **[3]**

9 The points $A(1, 2)$, $B(5, k)$ and $C(k, 2k)$ are three distinct collinear points, where k is a constant.

(a) Write down an expression, in terms of k, for the gradient of the line segment AB.

..... **[1]**

(b) By finding an expression for the gradient of BC, show that $k^2 - 11k + 10 = 0$.

[3]

(c) Solve the equation to find the two possible values of k.

$k =$ ..... , k =

..... **[2]**

(d) Calculate the exact gradient of the line passing through point B and the origin $(0, 0)$.

..... **[2]**

10 The points A, B and C have coordinates $(1, 3)$, $(4, 9)$ and $(8, y)$ respectively.

(a) The points A, B and C lie on the same straight line.
Work out the value of y.

$y =$ ..... **[2]**

(b) A different point D has coordinates $(p, p^2 - p - 3)$.
The gradient of the line AD is 2.
Work out the possible values of p.

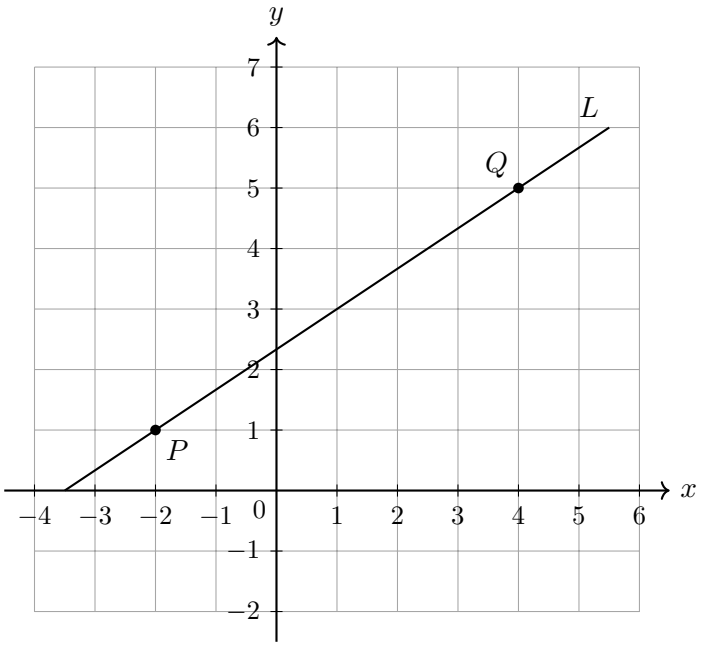
$p =$ ..... , $p =$ ..... **[4]**

(c) A line L has gradient m.
The line L passes through the points $(3, m)$ and $(m + 1, 12)$.
Find the two possible values of m.

$m =$ ..... , $m =$ ..... **[4]**

11 **(a)** The diagram shows a straight line L passing through the points $P(-2, 1)$ and $Q(4, 5)$.

Find the gradient of line L.



..... **[2]**

(b) A different line passes through the points $(3, k)$ and $(8, -2)$.

The gradient of this line is -1.4.

Calculate the value of k.

$k =$ ..... **[3]**

12 Point A has coordinates $(-3, 5)$ and point B has coordinates $(4, -9)$.

(a) Calculate the gradient of the line AB.

..... **[2]**

(b) Point C has coordinates $(k, 1)$.
The gradient of the line AC is equal to the gradient of the line AB.
Find the value of k.

$k =$ ..... **[3]**

13 Point P has coordinates $(k-3, 2k)$ and point Q has coordinates $(2, 5)$.

(a) The gradient of the line PQ is -3.
Work out the value of k.

$k =$ ..... **[3]**

(b) Using your value of k from part (a), find the coordinates of the midpoint of PQ.

(..... , ...) **[2]**

14 A straight line passes through the points $P(3k, 4)$ and $Q(k-2, 8)$.
The gradient of this line is $-\frac{1}{2}$.

(a) Find the value of k.

$k =$ ..... **[3]**

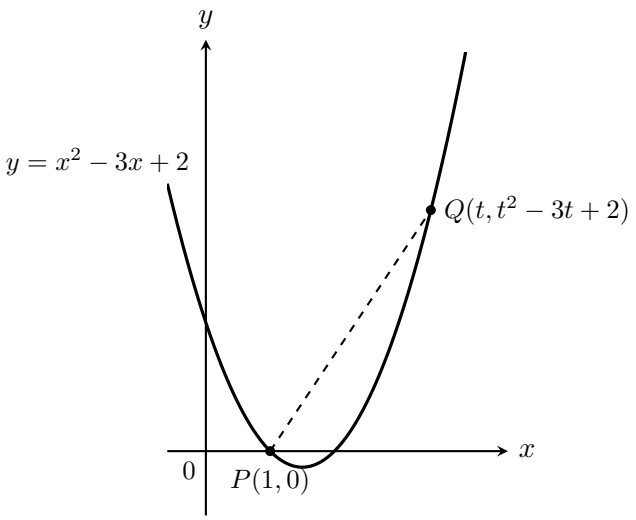
(b) Write down the coordinates of P and Q.

$P =$ ..... , $Q =$ ..... **[1]**

(c) Another point R lies on the y-axis such that the gradient of the line passing through P and R is 1.
Find the coordinates of R.

(..... , ...) **[3]**

15 The diagram shows a sketch of the curve $y = x^2 - 3x + 2$ and two points, $P(1, 0)$ and $Q(t, t^2 - 3t + 2)$, on the curve, where $t > 1$.



(a) Show that the gradient of the chord PQ simplifies to $t - 2$.

[3]

(b) Find the coordinates of point Q when the gradient of the chord PQ is 2.8.

$Q = (\ldots\ldots\ ,\ \ldots\ldots)$ **[3]**

(c) Point R has coordinates $(3.2, -1.5)$.
Find the value of t, where $t > 1$, for which the gradient of QR is equal to the gradient of PQ.
Give your answer correct to 3 significant figures.

$t =$ ..... **[4]**

1 $\frac{5-(-3)}{6-2}$ or equivalent **[M1]**

2 **[A1]**

[Back to Question] *[Total Marks: 2]*

2 **(a)** Evidence of using $\frac{\text{change in } y}{\text{change in } x}$ with two correct points from the graph, e.g., $\frac{4-(-2)}{3-0}$ **[M1]**

2 **[A1]**

(b) $(0, -2)$ **[B1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** Uses gradient formula: $\frac{-4-8}{4-(-2)}$ or $\frac{8-(-4)}{-2-4}$ **[M1]**

-2 **[A1]**

(b) Identifies coordinates of two clear points on the line, e.g., $(0, 1)$ and $(4, 4)$ **[M1]**

$\frac{3}{4}$ or 0.75 **[A1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** Uses the gradient formula: $\frac{-5-7}{2-(-4)}$ **[M1]**

-2 **[A1]**

(b) Sets up the equation: $\frac{c-7}{0-(-4)} = -2$ **[M1]**

-1 **[A1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** Correct vertical and horizontal change seen, e.g. $\frac{6}{4}$ or $\frac{3}{2}$ **[M1]**

1.5 (or $\frac{3}{2}$) **[A1]**

(b) Correct vertical and horizontal change seen, e.g. $\frac{-3}{6}$ **[M1]**

-0.5 (or $-\frac{1}{2}$) **[A1]**

(c) $\frac{-9-7}{5-(-3)}$ **[M1]**

-2 **[A1]**

[Back to Question] *[Total Marks: 6]*

6 **(a)** $\dfrac{k-5}{4-(-2)} = -\dfrac{1}{2}$ **[M1]**

$k - 5 = -3$ **[M1]**

2 **[A1]**

(b) Gradient of $BC = \dfrac{-1-2}{10-4} = -\dfrac{1}{2}$ **[M1]**

Gradient of AB = gradient of BC and point B is common to both segments **[A1]**

(c) $\dfrac{(3p-1)-5}{p-(-2)} = 2$ **[M1]**

$3p - 6 = 2(p+2)$ **[M1]**

10 **[A1]**

[Back to Question] *[Total Marks: 8]*

7 **(a)** $\dfrac{(a+4)-3a}{5a-2a}$ or equivalent **[M1]**

$\dfrac{4-2a}{3a}$ **[A1]**

(b) States or implies gradient of PQ is $-\dfrac{1}{3}$ **[B1]**

$\dfrac{4-2a}{3a} = -\dfrac{1}{3}$ or $3(4-2a) = -3a$ **[M1]**

4 **[A1]**

(c) Gradient $AB = \dfrac{8-2}{3-(-1)}$ or $\dfrac{6}{4}$ or $\dfrac{3}{2}$ **[M1]**

$\dfrac{23-8}{c-3} = \dfrac{3}{2}$ or $\dfrac{23-2}{c-(-1)} = \dfrac{3}{2}$ **[M1]**

13 **[A1]**

[Back to Question] *[Total Marks: 8]*

8 **(a)** $\dfrac{(a+6)-3a}{4-(-2)}$ **[M1]**

$\dfrac{6-2a}{6}$ **[M1]**

$\dfrac{3-a}{3}$ or $1 - \dfrac{a}{3}$ **[A1]**

(b) $\dfrac{3-a}{3} = 3$ oe **[M1]**

$a = -6$ **[A1]**

(c) $P = (-2, -18)$ and $Q = (4, 0)$ **[M1]**

$\left(\dfrac{-2+4}{2}, \dfrac{-18+0}{2}\right)$ **[M1]**

$(1, -9)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

9 **(a)** $\frac{k-2}{4}$ **[B1]**

(b) Finds an expression for the gradient of BC (e.g., $\frac{k}{k-5}$) and equates to the gradient of AB: $\frac{k-2}{4} = \frac{k}{k-5}$ **[M1]**

Expands brackets correctly: $(k-2)(k-5) = 4k$

$k^2 - 7k + 10 = 4k$ [M1]

Rearranges to the given equation $k^2 - 11k + 10 = 0$ with no mathematical errors seen **[A1]**

(c) Factorises $(k-1)(k-10) = 0$ or uses the quadratic formula **[M1]**

1 and 10 **[A1]**

(d) Deduces that $k = 10$ must be used, as $k = 1$ would mean points A and C are the same, contradicting that they are distinct **[B1]**

Calculates the gradient of OB using $(5, 10)$: $\frac{10-0}{5-0} = 2$ **[B1]**

[Back to Question] *[Total Marks: 8]*

10 **(a)** Calculates the gradient of AB as $\frac{9-3}{4-1} = 2$ and equates to gradient of BC or AC, e.g., $\frac{y-9}{8-4} = 2$ **[M1]**

$y = 17$ **[A1]**

(b) Sets up the gradient equation: $\frac{(p^2-p-3)-3}{p-1} = 2$ **[M1]**

Simplifies to form a 3-term quadratic equation: $p^2 - p - 6 = 2(p-1)$

$p^2 - 3p - 4 = 0$ [M1]

Factorises $(p-4)(p+1) = 0$ or uses the quadratic formula correctly **[M1]**

$p = 4$ and $p = -1$ **[A1]**

(c) Sets up the gradient equation: $\frac{12-m}{(m+1)-3} = m$ **[M1]**

Multiplies by the denominator: $12 - m = m(m-2)$ **[M1]**

Rearranges into a 3-term quadratic: $m^2 - m - 12 = 0$ **[M1]**

$m = 4$ and $m = -3$ **[A1]**

[Back to Question] *[Total Marks: 10]*

11 **(a)** $\frac{5-1}{4-(-2)}$ or equivalent **[M1]**

$\frac{2}{3}$ **[A1]**

(b) $\dfrac{-2-k}{8-3} = -1.4$ or equivalent **[M1]**

$-2-k=-7$ **[M1]**

5 **[A1]**

[Back to Question] *[Total Marks: 5]*

12 (a) $\dfrac{-9-5}{4-(-3)}$ or $\dfrac{5-(-9)}{-3-4}$ **[M1]**

-2 **[A1]**

(b) $\dfrac{1-5}{k-(-3)} = -2$ oe **[M1]**

$-4=-2(k+3)$ or $-2k-6=-4$ **[M1]**

$k=-1$ **[A1]**

[Back to Question] *[Total Marks: 5]*

13 (a) Forms a correct equation for the gradient: $\dfrac{5-2k}{2-(k-3)} = -3$ OR $\dfrac{2k-5}{k-3-2} = -3$ **[M1]**

Correctly simplifies the denominator and equates: $5-2k=-3(5-k)$ **[M1]**

$k=4$ **[A1]**

(b) Evaluates coordinates of P as $(1,8)$ and substitutes into midpoint formula: $\left(\dfrac{1+2}{2}, \dfrac{8+5}{2}\right)$ **[M1]**

$(1.5, 6.5)$ **[A1]**

[Back to Question] *[Total Marks: 5]*

14 (a) Finds a correct algebraic expression for the gradient: $\dfrac{8-4}{k-2-3k}$ or $\dfrac{4}{-2k-2}$ **[M1]**

Equates to $-\dfrac{1}{2}$ and eliminates fractions: $8=2k+2$ or $-8=-2k-2$ **[M1]**

3 **[A1]**

(b) $P(9,4)$ and $Q(1,8)$ **[B1]**

(c) Identifies the coordinates of R must be of the form $(0,y)$ **[B1]**

Sets up the gradient equation: $\dfrac{y-4}{0-9} = 1$ **[M1]**

$(0,-5)$ **[A1]**

[Back to Question] *[Total Marks: 7]*

15 (a) Gradient $= \dfrac{(t^2-3t+2)-0}{t-1}$ **[M1]**

Factorising numerator: $\dfrac{(t-1)(t-2)}{t-1}$ **[M1]**

Cancelling $(t-1)$ correctly to obtain $t-2$ **[A1]**

(b) $t-2=2.8$ giving $t=4.8$ **[M1]**

$y=4.8^2-3(4.8)+2$ **[M1]**

$(4.8, 10.64)$ **[A1]**

(c) Gradient of $QR=\dfrac{(t^2-3t+2)-(-1.5)}{t-3.2}$ **[M1]**

$\dfrac{t^2-3t+3.5}{t-3.2}=t-2$ **[M1]**

$t^2-3t+3.5=t^2-5.2t+6.4$ leading to $2.2t=2.9$ **[M1]**

1.32 **[A1]**

[Back to Question] *[Total Marks: 10]*